

Challenge 7: The Sphere Packing Problem

Definition 1. Fix a dimension $r \in \mathbb{N}$. Consider the vector space $\mathbb{R}^r$, equipped with Euclidean distance and Lebesgue measure $\text{Vol}(\cdot)$. For $x \in \mathbb{R}^r$ and $\varepsilon > 0$, let $B_r(x, \varepsilon)$ denote the open ball in $\mathbb{R}^r$, with center x and radius ε .

1. A *sphere packing* of $\mathbb{R}^r$ is a set C of points in $\mathbb{R}^r$ that have pairwise distance at least two.
2. The *upper density* of a sphere packing C is

$$\bar{\delta}_r(C) := \limsup_{R \rightarrow \infty} \frac{\text{Vol}(\bigcup_{x \in C} B(x, 1) \cap B(0, R))}{\text{Vol}(B(0, R))}.$$

The *sphere-packing constant* is $\Delta_r := \sup_C \bar{\delta}_r(C)$ where the supremum ranges over all sphere packings C .

3. A *lattice packing* of $\mathbb{R}^r$ is a sphere packing C which is also a lattice in $\mathbb{R}^r$, i.e., $C = \{\sum_{i=1}^r a_i v_i : a_1, \dots, a_r \in \mathbb{Z}\}$ for some basis $\{v_1, \dots, v_r\}$ of $\mathbb{R}^r$. The *lattice-packing constant* is $\Delta_r^L := \sup_C \bar{\delta}_r(C)$ where the supremum ranges over all lattice packings of $\mathbb{R}^r$.

The sphere packing problem is to determine Δ_r . It is not even known whether densest sphere packings always need to be attained by lattices:

Open Question 1.1. Given $r \geq 1$, is it true that $\Delta_r = \Delta_r^L$?

In the literature, the equation $\Delta_r = \Delta_r^L$ is only shown for $r \in \{1, 2, 3, 8, 24\}$. The case $r = 3$ is known as the Kepler conjecture, proven by Hales [1]. The cases $r = 8$ and $r = 24$ were settled by Viazovska [2] and by Cohn, Kumar, Miller, Radchenko and Viazovska [3], respectively.

References

- [1] T. C. Hales. “A proof of the Kepler conjecture”. In: *Annals of Mathematics* 162.3 (2005), pp. 1065–1185. DOI: 10.4007/annals.2005.162.1065.
- [2] M. S. Viazovska. “The sphere packing problem in dimension 8”. In: *Annals of Mathematics* 185.3 (2017), pp. 991–1015. DOI: 10.4007/annals.2017.185.3.7.
- [3] H. Cohn et al. “The sphere packing problem in dimension 24”. In: *Annals of Mathematics* 185.3 (2017), pp. 1017–1033. DOI: 10.4007/annals.2017.185.3.8.